

Developing Good Research Habits

bit.ly/research_habits

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Outline

- 1 Citing
- 2 Searching
- 3 Writing
- 4 Tracking

Alternative titles

There's more to being a
researcher than writing a thesis

Alternative titles

There's more to being a
researcher than writing a thesis

What you should know but
probably don't

Alternative titles

There's more to being a researcher than writing a thesis

What you should know but probably don't

Listen up, young padawans



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Managing references

Zotero

- ➔ Free and on all operating systems
- ➔ Web-version and local version synced
- ➔ Browser extension for adding papers/books
- ➔ Attach notes and annotations to papers.
- ➔ Works with Word, LibreOffice or LaTeX.
- ➔ Generate bibliography automatically
- ➔ Handles all formatting for you.

zotero

To install:

- Set up account at www.zotero.org
- Download from www.zotero.org

Managing references

Paperpile

- ➔ \$3 per month and runs on Google Chrome
- ➔ Papers stored on Google Drive
- ➔ Browser extension for adding papers/books
- ➔ Works with Google Docs or LaTeX.
- ➔ Generate bibliography automatically
- ➔ Handles all formatting for you.
- ➔ Amazingly fast



To install:

- Set up account at paperpile.com
- Download Google chrome browser extension

What to cite?

- Cite what is important.
- Cite (only) what is relevant.
- Avoid lists of gratuitous references.
- Include proper citations for all packages and software you use.



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When using R

```
citation("packagename")
```



Sight what you cite

- Every article cited should be sighted, & preferably read.
- At the very least, check that the article cited really does say what you think it says.
- Don't just cite what other people say about citations.
- Store accurate reference info from the start.
- Give credit where it is due.



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Google Scholar



☒ Articles ☐ Case law

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A free, AI-powered research tool for scientific literature

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Reproducibility

Not reproducible:

- Data edited in a spreadsheet
- Click and point analysis
- Copy and paste graphs and tables
- Tables typed by hand

Reproducible

- All data edits scripted
- All analysis scripted
- Graphs and tables automatically pulled in to the thesis
- Tables generated with scripts



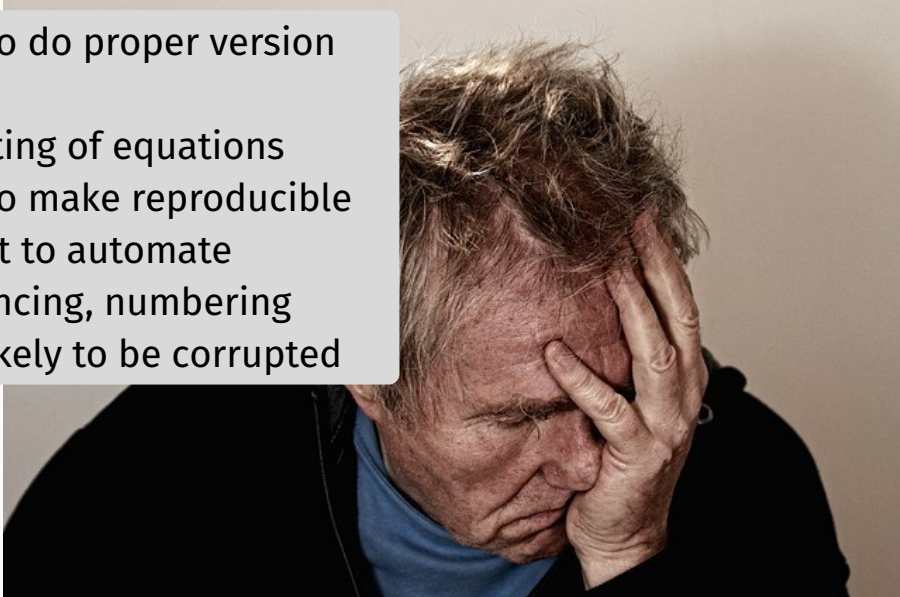
**STOP CLICKING
AND START SCRIPTING**

Microsoft Word



Microsoft Word

- Impossible to do proper version control
- Poor formatting of equations
- Impossible to make reproducible
- More difficult to automate cross-referencing, numbering
- Files more likely to be corrupted





My Honours Thesis

Student Name

A thesis submitted for the degree of
Bachelor of Commerce (Honours)
at Monash University in 2026
Department of Econometrics & Business Statistics

Supervised by: Prof. Supervisor Name

Quarto

- ➔ Combines R and Python code with text in a single file
- ➔ Reproducible research
- ➔ Monash Thesis Honours Template via <https://GitHub.com/quarto-monash/honours-thesis>
- ➔ Useful for assignments too
- ➔ See quarto.org for help

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Version control

- `thesis_v1`, `thesis_v2`, etc., is not adequate version control.
- You need to track changes over time, have a *remote* repository, and be able to roll back as required.
- Your repository should contain *everything* required to produce your thesis including computer code, references, writing.
- Your repository should have an obvious structure and be fully documented.
- **GitHub** solves these problems
- Read “Happy git with R”: happygitwithr.com



Version control with git

- RStudio integrates with GitHub, so version control built in.
- But GitHub can be used with *any* text-based language including Stata, Python, LaTeX, R, Rmarkdown, Quarto, markdown, etc.
- Git allows you to:
 - ▶ track changes
 - ▶ experiment in branches
 - ▶ undo
- GitHub provides:
 - ▶ backup and restore
 - ▶ synchronisation



Slides and links:

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